

MULTIMEDIA UNIVERSITY OF KENYA

DSA ASSIGNMENT

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PROGRAM: SOFTWARE ENGINEERING

PARKING SYSTEM ASSIGNMENT

UNIT: DATA STRUCTURES AND ALGORITHM

Algorithm for the Smart Parking Management System

The system is broken down into six modules, matching the client's requirements: slot display, vehicle entry, fee calculation, payment and barrier control, rate configuration, and reporting. Each module's algorithm is written as simple step-by-step pseudocode below, with a short explanation of the logic behind each step.

Module 1: Slot Display / Availability Module

Purpose: Show live slot status on a display board / web page so drivers know availability before entry.

START

1. Initialize 20 slots, each marked as FREE
2. LOOP continuously (or on page refresh):
 - a. Count number of slots marked FREE
 - b. Display each slot number with its status (FREE / OCCUPIED)
 - c. Show total free slots count

END

Explanation: Step 1 sets up the initial state of the parking lot before any vehicle arrives. Step 2 repeats so the display always reflects the current state — this is why it is a loop rather than a one-time check. Counting free slots (2a) gives management a quick summary figure, while listing each slot individually (2b) lets a driver see exactly which bays are open. This module does not change any data itself; it only reads the current status set by Modules 2 and 4.

Module 2: Vehicle Entry Module

Purpose: Record a vehicle on arrival and allocate it a bay automatically.

START

1. Driver arrives, enters number plate
2. Check if any slot is FREE
3. IF no slot is FREE:
 - Display "Parking Full" message
 - STOP
4. ELSE:

- a. Find first available FREE slot
- b. Record: number plate, entry time (current time), allocated slot number
- c. Mark that slot as OCCUPIED
- d. Save record to vehicle list/database
- e. Display allocated slot number to driver

END

Explanation: Step 2 is a guard condition that prevents the system from allocating a bay that does not exist — this handles the exception case of a full parking lot before doing any allocation work. Step 4a searches for the first free slot, which is a simple and fair allocation strategy for a system of this size (20 slots). Step 4b captures the three pieces of data the client specifically asked for: plate number, time of entry, and allocated bay. Marking the slot OCCUPIED (4c) immediately updates Module 1's display, and saving the record (4d) makes the data available later for Module 3 when the vehicle exits.

Module 3: Duration and Fee Calculation Module

Purpose: Automatically compute how long a vehicle has been parked and what it owes, at the point of exit.

START

1. Driver enters number plate at exit
2. Search vehicle records for matching plate
3. IF plate not found:
 - Display "Record not found"
 - STOP
4. ELSE:
 - a. Get entry time from record
 - b. Get current time as exit time
 - c. Calculate duration = exit time - entry time
 - d. Determine fee based on duration:
 - IF duration <= 30 minutes: fee = 0
 - ELSE IF duration <= 2 hours: fee = 50
 - ELSE IF duration <= 4 hours: fee = 100
 - ELSE IF duration <= 6 hours: fee = 300
 - ELSE: fee = 500
 - e. Display duration and fee to driver

END

Explanation: Step 2 links this module back to the record created in Module 2 using the number plate as the search key, which is why the plate must be unique per active vehicle. Step 3 handles the exception where a driver tries to exit without a valid entry record. Steps 4a-4c calculate elapsed time using simple subtraction of timestamps. Step 4d applies the client's tiered fee structure using a sequence of IF/ELSE IF conditions checked from the smallest duration upward, ensuring each vehicle falls into exactly one bracket. This tiered structure means rates can later be changed (Module 5) without altering this decision logic.

Module 4: Payment and Barrier Control Module

Purpose: Collect payment and only open the barrier once payment is confirmed.

START

1. Display fee amount and payment options (M-Pesa, Card, Cash)
2. Driver selects payment method and confirms payment
3. IF payment confirmed:
 - a. Mark transaction as PAID
 - b. Free up the slot (mark as FREE)
 - c. Remove/close vehicle record (move to transaction log)
 - d. Open barrier (simulate: display "Barrier Open")
4. ELSE:
 - Keep barrier closed
 - Display "Payment pending"

END

Explanation: This module enforces the client's requirement that the barrier must only open on confirmed payment (step 3 vs step 4 is the enforcement point). Freeing the slot (3b) updates Module 1's display in real time, so the bay becomes visible as available again immediately. Moving the record to a transaction log (3c) rather than deleting it preserves the data for Module 6's audit trail, satisfying the requirement for a reconciliation and VAT record.

Module 5: Rate Configuration Module

Purpose: Allow management to change parking rates at any time without modifying the program's code.

START

1. Admin opens rate settings page
2. Display current rates (30 min, 2hr, 4hr, 6hr, over 6hr)
3. Admin enters new rate values
4. Save updated rates to rate configuration (separate from main code)
5. All future fee calculations use the new rates

END

Explanation: Storing rates separately from the program logic (step 4) is the key idea here — Module 3 reads the current rate values from this configuration rather than having them fixed in the code, so step 5 works automatically without a software change. This directly satisfies the client's requirement to adjust rates at any time.

Module 6: Reporting / Audit Module

Purpose: Keep an auditable record of every shilling collected, for reconciliation and VAT.

START

1. On every completed exit transaction, record:
 - plate number, entry time, exit time, duration, amount paid, payment method
2. Append record to transaction log (database/file)
3. Admin can view full log anytime for reconciliation and VAT purposes

END

Explanation: Appending each completed transaction (step 2) rather than overwriting data builds a permanent, chronological log. Capturing all six data points (step 1) gives management everything needed to reconcile daily cash collected against system records and to support VAT reporting, which was an explicit client objective.

Overall Flow

Module 2 (Entry) feeds Module 1 (Display) by changing slot status. On exit, Module 3 (Fee Calculation) reads both the vehicle record from Module 2 and the current rates from Module 5. Module 4 (Payment and Barrier) acts on Module 3's output and, once payment succeeds, updates Module 1's display again and writes into Module 6's log. This flow ensures every requirement in the client's terms of reference is covered by exactly one module, with no overlaps.